Reasoning and Learning in Artificial Intelligence

5.1 Reasoning

Reasoning is the process of analyzing available information, comparing it with existing knowledge, and drawing conclusions.

It helps in:

- Decision making
- Problem solving
- Prediction
- Evaluation of actions

Example

Virtual assistants such as Siri use reasoning to answer questions, provide directions, and make recommendations.

5.1.1 Features of a Reasoning System

A good reasoning system has the following characteristics:

1. Self-Contained System

- Works independently.
- Does not require constant modification by programmers.

2. Solves Different Problems

- Can learn through examples, instructions, or trial-and-error.
- Can solve new and unfamiliar problems.

3. Improves with Experience

- Performance improves after a limited number of trials.

4. Learns Quickly

- Can solve new problems using only a small number of examples.

5.1.2 Why Focus on Reasoning Systems?

Reasoning systems help machines:

- Analyze information.
- Solve problems.
- Make decisions.
- Draw conclusions.

Difference Between Prediction, Inference and Reasoning

Prediction	Inference	Reasoning
Uses pattern matching	Generates new information	Goal-oriented process
Maps inputs to outputs	Automatic process	Produces desired conclusions
Example: Weather forecast	Deriving facts	Decision making

5.1.3 Methods of Reasoning

The major reasoning methods are:

1. Deductive Reasoning
2. Inductive Reasoning
3. Common Sense Reasoning
4. Monotonic Reasoning
5. Non-Monotonic Reasoning
6. Abductive Reasoning

1. Deductive Reasoning

Definition

Deductive reasoning uses general facts or rules to reach specific conclusions.

Approach

Top-Down Approach

General Rule → Specific Conclusion

Example

Premise 1: All humans breathe.

Premise 2: Naina is a human.

Conclusion: Naina breathes.

Characteristics

- Based on facts and rules.
- Produces logically certain conclusions.
- Conclusion is true if premises are true.

Advantages

- Highly reliable.
- Gives accurate conclusions.

Disadvantages

- Requires true facts.
- Difficult to apply in uncertain situations.

2. Inductive Reasoning

Definition

Inductive reasoning uses specific observations to form general conclusions.

Approach

Bottom-Up Approach

Specific Facts → General Rule

Example

- All students scoring 100% are from State X.
- Therefore, students from State X are highly talented.

Characteristics

- Generates hypotheses.
- Conclusion may not always be true.
- Exploratory in nature.

Advantages

- Works with limited data.
- Fast and flexible.

Disadvantages

- Conclusions may be incorrect.
- Less certain than deductive reasoning.

Difference Between Deductive and Inductive Reasoning

Deductive Reasoning	**Inductive Reasoning**
Top-down approach	Bottom-up approach
General to specific	Specific to general
Produces certain conclusions	Produces probable conclusions
Requires facts	Requires evidence

3. Common Sense Reasoning

Definition

Reasoning based on everyday experiences and practical knowledge.

Examples

- Wearing a mask during a lockdown to avoid fines.
- Looking both ways before crossing a road.

Characteristics

- Informal reasoning.
- Based on experiences.

- Uses good judgment.
- Uses heuristic knowledge.

Applications

- Computer Vision
- Robotics
- Intelligent Assistants

Advantages

- Mimics human thinking.
- Useful in real-world situations.

Disadvantages

- Not always logically accurate.
- Depends on personal experiences.

4. Monotonic Reasoning

Definition

Once a conclusion is derived, it never changes even if new information is added.

Examples

- The Harappan Civilization existed in the Indus Valley.
- Kashmir is famous for natural beauty.

Characteristics

- Conclusions remain fixed.
- New facts do not affect old conclusions.
- Used in logic-based systems.

Advantages

1. Proofs remain valid.
2. Derived facts remain valid forever.

Disadvantages

1. Cannot handle real-world uncertainty.
2. Cannot revise conclusions.
3. Cannot represent hypothetical knowledge.

5. Non-Monotonic Reasoning

Definition

A conclusion can change when new information becomes available.

Example

Knowledge Base:

- Birds fly.
- Tweety is a bird.

Conclusion:

- Tweety flies.

Later information:

- Tweety is a penguin.

New Conclusion:

- Tweety does not fly.

Characteristics

- Handles incomplete information.
- Supports changing conclusions.
- Represents real-world situations.

Advantages

1. Suitable for real-world applications.
2. Supports assumptions and probabilities.

Disadvantages

1. Old conclusions may become invalid.
2. Not suitable for theorem proving.

6. Abductive Reasoning

Definition

Abductive reasoning starts with observations and finds the most likely explanation.

Example

A doctor diagnoses a disease using current test reports.

Characteristics

- Uses present observations.
- Finds the most probable explanation.
- Useful for decision-making and troubleshooting.

Advantages

- Useful in diagnosis.
- Helps make decisions under uncertainty.

Disadvantages

- Conclusions may not always be correct.
- Can justify incorrect beliefs.

5.2 Probabilistic Reasoning in AI

Probabilistic Reasoning is a technique that combines **probability theory** with **logic** to handle uncertainty in Artificial Intelligence.

Why is Probabilistic Reasoning Needed?

In real-world situations:

- Information may be incomplete.
- Facts may be uncertain.
- Outcomes may be unpredictable.

- Errors may occur during observations.

Example

"It may rain today."

We cannot say with complete certainty whether it will rain or not. Therefore, probability is used to express such uncertainty.

5.2.1 Uncertainty

Definition

Uncertainty occurs when we are not completely sure whether a statement or event is true or false.

Causes of Uncertainty

1. Unreliable information sources
2. Experimental errors
3. Equipment faults
4. Temperature variations
5. Climate changes

Example

A weather forecast predicts a 70% chance of rain. Rain may or may not occur, creating uncertainty.

5.2.2 Probability

Definition

Probability is the measure of the likelihood that an event will occur.

Range of Probability

$0 \leq P(A) \leq 1$

Where:

- $P(A) = 0 \rightarrow$ Event will never occur (Impossible Event)

- P(A) = 1 → Event will definitely occur (Certain Event)

Example

- Probability of the sun rising tomorrow = 1
- Probability of rolling a 7 on a six-sided die = 0

Complement Rule

The probability of an event occurring and not occurring adds up to 1.

Formula

Where:

- P(A) = Probability of event A occurring
- P(¬A) = Probability of event A not occurring

Example

If probability of rain is 0.7,

Then:

P(¬Rain) = 1 - 0.7 = 0.3

Important Terms in Probability

1. Event

Definition

An event is a possible outcome of an experiment.

Example

Coin Toss:

- Head
- Tail

Each outcome is called an event.

2. Sample Space

Definition

The set of all possible outcomes of an experiment.

Example

For a coin toss:

S = {Head, Tail}

For rolling a die:

S = {1, 2, 3, 4, 5, 6}

3. Prior Probability

Definition

Probability calculated before observing any new information.

Example

Before checking the weather report:

Probability of rain = 0.4

This is Prior Probability.

4. Posterior Probability

Definition

Probability calculated after receiving new evidence or information.

Example

After seeing dark clouds:

Probability of rain = 0.8

This updated probability is called Posterior Probability.

5. Independent Events

Definition

Events that do not affect each other.

Example

Tossing a coin twice.

The result of the first toss does not affect the second toss.

Formula

$P(A|B) = P(A)$

Easy Example

- First Toss = Head
- Second Toss = Tail

Both events are independent.

6. Dependent Events

Definition

Events whose outcomes depend on previous events.

Example

A bag contains:

- 2 Blue balls
- 3 Red balls

If one ball is removed without replacement, the next draw depends on the first draw.

7. Marginal Probability

Definition

Probability of a single event occurring without considering any other event.

Example

Probability of drawing a red card from a deck:

P(A) = 26/52 = 0.5

Meaning:

There is a 50% chance of drawing a red card.

8. Joint Probability

Definition

Probability of two events occurring together.

Formula

P(A ∩ B)

Example

Probability of drawing a card that is:

- Red
- Number 8

P(A ∩ B) = 2/52 = 1/26

9. Conditional Probability

Definition

Conditional probability is the probability of an event occurring when another event has already occurred.

Formula

Where:

- P(A∩B) = Joint Probability
- P(B) = Probability of Event B

Easy Example

Suppose:

- 30 students like both Cricket and Football.
- 65 students like Football.

Find the probability that a student likes Cricket given that the student likes Football.

$P(A \cap B)=30/100=0.30$
$P(B)=65/100=0.65$

$P(A|B)=0.30/0.65$
≈ 0.46

Therefore,

There is a 46% probability that a student likes Cricket given that the student likes Football.

Solved Example

Question

70% students want to learn R language.

40% students want to learn Python.

Find the percentage of students who like both R and Python.

Solution

Let:

- A = Students liking Python
- B = Students liking R

Given:

$P(A|B)=0.40$
$P(B)=0.70$

Using:

$P(A \cap B)=P(A|B) \times P(B)$

$P(A \cap B)=0.40 \times 0.70$
$=0.28$

Answer

$0.28 \times 100 = 28\%$

Therefore, 28% of students like both R and Python.